LIFELINE-RL: A BENCHMARK FOR TOPOLOGY-AWARE AND HISTORY-DEPENDENT SELF-PLAY

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ABSTRACT

We introduce LIFELINE-RL, a deterministic self-play benchmark combining a triangular lattice, player-specific logical connectivity, cascading removal, and history-dependent repetition. Complete side-five enumeration finds reachable latent topology differences but no difference in legal actions, transitions, exact values, or optimal actions; a standard-start side-six replay instead shows Superko preventing a repeatable six-transition loop. We compare a padded CNN, physical-edge Grid-GNN, and typed-edge Topology-GNN in 15 matched mixed-size runs over sides 5, 7, and 9. All three beat Random at every seen size. Scores against the fixed UCT-MCTS-16 opponent vary by model and size and are reported only as pooled descriptive evidence. Topology-GNN outscores Grid-GNN in direct play at all three sizes, but each 200-game task repeats only two deterministic trajectories; the evidence does not support a global representation ranking. All empirical claims are bounded to frozen receipts and task-indexed replay-verified game records.

1 INTRODUCTION

Most board-game benchmarks expose local occupancy on a fixed lattice. LIFELINE-RL instead separates rendered occupancy from player-specific logical connectivity: moves can cut edges, remove disconnected components, and invoke a repetition rule whose legality depends on history. We ask which representations support self-play under these mechanisms.

Our contributions, each tied to an auditable artifact and an explicit boundary, are:

1. a tested variable-size environment with exact snapshots, legal masks, and replayable trajectories, supported by rule-engine parity traces but not an exhaustive cross-implementation proof;
2. exhaustive side-five and bounded larger-board diagnostics separating latent topology aliasing from decision-relevant aliasing and Superko activation; and
3. a frozen five-seed comparison of three representations backed by 330 receipts and 30,600 task-indexed replay-verified game records, without a global model ranking.

The environment contribution does not depend on a positive model result. We claim neither non-Markov observations, relative sample efficiency, nor unseen-size generalization without a corresponding gate-passing formal result.

Claim control. A frozen claim–evidence–limitation ledger accompanies the draft. Every primary claim names its authoritative CSV/JSON or verifier and retains the relevant scope restriction; failed and inconclusive cells remain in all tables.

2 RELATED WORK

Self-play and game environments. AlphaGo Zero and AlphaZero established policy–value learning with search from self-play, while expert iteration, ELF OpenGo, and MuZero broadened the algorithmic and reproducibility picture (Silver et al., 2017; 2018; Anthony et al., 2017; Tian et al., 2019; Schrittwieser et al., 2020). General-purpose multi-agent and game testbeds include OpenSpiel

and PettingZoo (Lanctot et al., 2019; Terry et al., 2020); MCTS is surveyed by Browne et al. (2012). LIFELINE-RL is narrower than these platforms but targets a particular confound: rendered occupancy, player-specific logical connectivity, and repetition history can be distinct parts of simulator state.

General game systems and structural transfer. Ludii and Regular Boardgames provide complementary declarative systems for representing and executing board-game rules, including dedicated reasoning work for the latter (Piette et al., 2020; Kowalski et al., 2019; 2020); Polygames supplies zero-learning infrastructure across multiple games (Cazenave et al., 2021). Prior work transfers policy–value networks across Hex sizes, games, and variants (Gao et al., 2019; Soemers et al., 2021), and evaluates graph representations directly in chess RL and DQN-based Hex agents (Rigaux & Kashima, 2024; Keller et al., 2023). These studies are the closest empirical neighbors, but do not establish topology-input superiority in LIFELINE-RL or unseen-size transfer for our models.

Graph representations and size. Message passing, inductive graph learning, relational biases, and graph reinforcement learning motivate explicit structural inputs (Gilmer et al., 2017; Hamilton et al., 2017; Battaglia et al., 2018; Jiang et al., 2018). Work on neural algorithm execution, combinatorial optimization, and TSP generalization shows why graph structure alone does not justify extrapolation claims (Xu et al., 2019; Joshi et al., 2020). We therefore compare representations only on jointly trained sizes and reserve unseen-size evaluation for future formal artifacts.

State sufficiency, masks, and repetition. Partial observability, perceptual aliasing, and state abstraction formalize the risks of discarding history or decision-relevant variables (Kaelbling et al., 1998; Chrisman, 1992; Givan et al., 2003; Abel et al., 2017). Invalid-action masking further shows that legality information is part of the learning interface (Huang & Ontañón, 2022). Go's legality, graph-history interaction, combinatorics, and legal-position counting provide closely related precedent for repetition-sensitive state accounting (Kishimoto & Müller, 2004; Tromp & Farnebäck, 2007; Tromp, 2016). Our diagnostics accordingly distinguish a full simulator state, an implemented masked observation, and deliberately weaker projections.

3 THE LIFELINE-RL BENCHMARK

3.1 BOARD AND ACTIONS

A board of side length n contains

$$N(n) = \frac{n(n+1)}{2} \tag{1}$$

valid lattice points. A turn selects one lattice point or the pass action, for $N(n) + 1$ discrete actions. The main benchmark uses two players and canonicalizes the observation to the player to act.

3.2 PHYSICAL OCCUPANCY AND LOGICAL TOPOLOGY

Let B_t denote physical point occupancy and let $E_t^{(b)}, E_t^{(w)}$ denote the explicit logical edge sets of the two players. Physical line points are used for rendering and blocking, whereas logical edges define connectivity to the protected initial node. An attack can break an edge and remove nodes and line points that are no longer logically connected.

3.3 FULL SIMULATOR STATE

We model the full state as

$$x_t = \left(B_t, E_t^{(b)}, E_t^{(w)}, H_t, p_t, c_t, z_t\right), \tag{2}$$

where H_t is the visited-state set used by the repetition rule, p_t is the player to act, c_t is the consecutive-pass count, and z_t is the terminal flag and cached terminal outcome. Resignation is outside the frozen training environment.

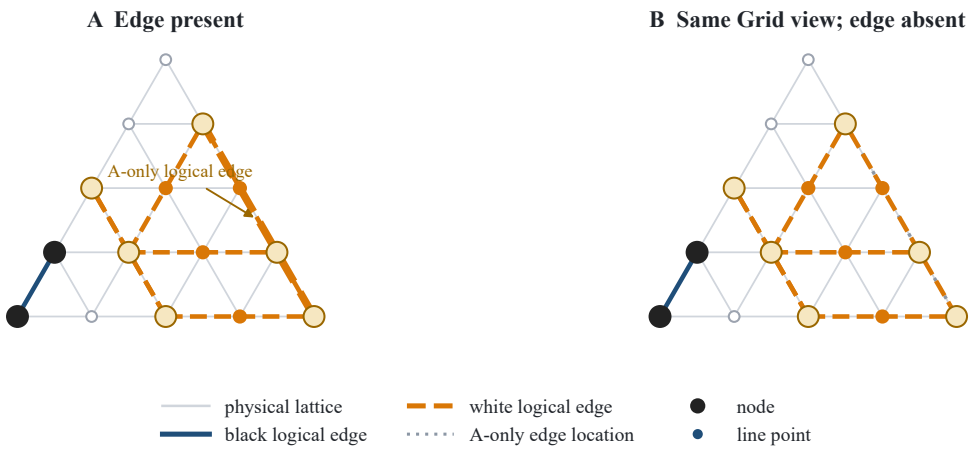


Figure 1: A replay-verified state pair reconstructed from the frozen side-five witness ledger. Both reachable states have identical occupancy, player to act, pass count, and legal mask; only history A contains the highlighted white logical edge. The dotted segment in history B marks that edge's absent location. This demonstrates latent topology aliasing only: the exhaustive side-five audit found no decision-relevant consequence.

Table 1: Pre-registered observation variants from the frozen benchmark protocol. The simulator always retains the exact full state; the table does not imply that any exposed feature is independently necessary.

Name	Exposed structure	Purpose
Grid	Occupancy and turn features	Flat/spatial baseline
GridGraph	Grid plus fixed lattice edges	Size-agnostic graph baseline
Topology	GridGraph plus typed logical edges	Main reference model
TopologyHistory	Topology plus compact history features	History ablation

3.4 Observation protocols

All four protocols also expose the player to act, consecutive-pass count, and legal-action mask. Thus a legal-set difference alone is not an alias under the implemented Grid protocol; it is an alias only for a separately named mask-free occupancy projection.

3.5 Rules and terminal reward

The executable rules distinguish occupied nodes, physical line points, and logical edges. Point actions extend or attack connectivity subject to protected regions and the three-point restriction; attacks can cut edges and cascade through components disconnected from a player's protected root. Superko transactionally rejects repeated canonical positions. Passing and automatic skips use the exact consecutive-pass terminal rule, after which discrete territory determines the winner. Training uses only terminal win/draw/loss reward $(+1, 0, -1)$.

3.6 Implementation and validation

The dependency-free Python environment uses a fixed point ordering followed by PASS, lossless cloning and versioned serialization, legal-action masks, and replayable action sequences. The frozen environment/search suite contains 62 passing tests. An expanded repository discovery snapshot on 2026-08-25 ran 88 tests: 81 passed and seven optional PyTorch-dependent tests were skipped because the dependency was unavailable; skips are not counted as failures. Five targeted and 60 seeded random trajectories match the independent Web rule engine step by step across side lengths 5, 7, 9, 10, 12, and 15. Targeted tests assert exact component deletion, edge-only cuts, edge restoration, and

transactional Superko rollback. Median end-to-end throughput in a three-repeat local engineering benchmark ranges from 1,574 transitions/s at side length 5 to 19.3 transitions/s at side length 15. These measurements include legal-mask and observation construction and are not optimized simulator-only timings.

4 STATE-ALIASING DIAGNOSTICS

Definition 1 (Implemented Grid equivalence)**.** *Two full states x and x' are Grid-equivalent if their implemented Grid observations are identical: physical occupancy, fixed coordinate features, player to act, consecutive-pass count, and legal-action mask all agree. Logical edges and exact repetition histories may differ.*

We use *occupancy projection* for the deliberately weaker map that omits the action mask. Results for that projection are never reported as aliases of the implemented Grid observation.

Definition 2 (Decision-relevant aliasing)**.** *An observation map ϕ aliases two states in a decision-relevant manner if $\phi(x) = \phi(x')$ but the states differ in legal actions, observable transition outcomes, returns, or optimal actions under a fixed game solution concept.*

Proposition 1 (Reachable topology aliasing: computational witness)**.** *For side length $n = 5$, there exist two legal histories that reach states with identical implemented Grid observations but different logical edge sets.*

Proposition 2 (Side-five negative result: exhaustive computation)**.** *Under the frozen two-player rules and standard side-five start, no pair of reachable Grid-equivalent states differs in legal actions, next Grid observation, exact zero-sum value, shared-action Q value, or optimal-action set. No reachable side-five move is rejected by Superko.*

Evidence and boundary. The complete raw graph contains 25,096 states and 67,505 transitions and is a DAG. It contains 976 Grid-alias groups and 1,088 pairs with different latent logical topology. An independent all-PASS backward-induction audit finds zero strict differences across every pair. This proves a side-five negative result, not nonexistence at sizes 6–15. The replay witness therefore establishes weak full-state aliasing only; the paper does not currently claim that Grid is non-Markov or strategically insufficient across the benchmark family.

Separately, a standard-start side-six trajectory naturally reaches a move that is rejected only by Superko. If enforcement is disabled while repetitions are still logged, that move closes a six-transition loop that can repeat. This establishes that the rule can bind, but not the stricter paired-state claim: we have not found two naturally reachable histories with identical mask-free Topology and different legality for a shared action.

Witness construction. Complete enumeration is combined with seeded random walks, collision indexing, and bounded shared-continuation search. Versioned witnesses retain legal histories, observation digests, full-state fingerprints, and replay assertions; unsuccessful larger-board searches retain their budgets and are labelled `NOT_FOUND_WITHIN_BUDGET`.

5 BENCHMARK PROTOCOL

5.1 SEARCH BASELINES

Random, Greedy, depth-2/3 Minimax, and UCT-MCTS share one color-balanced, replayable arena. Random samples uniformly from all legal point moves and PASS; Greedy is a deterministic one-ply heuristic. Minimax retains at most 20 tactically ordered point candidates plus PASS and is therefore approximate. UCT is adversarial, stores values from the root-player perspective, and uses uniform-random rollouts; the frozen UCT-MCTS-16 opponent uses 16 simulations per move, exploration constant $\sqrt{2}$, and rollout depth 80. It selects by visit count, then mean value, with seeded random tie-breaking; unterminated rollouts at the depth cap return zero. The simulator handles PASS, automatic skips, terminal scoring, and Superko for every baseline. The complete five-agent matrix remains an engineering smoke test; formal neural evaluation uses Random and UCT-MCTS-16, matched only on the 16-simulation search budget.

5.2 NEURAL SELF-PLAY

All models use the same AlphaZero-style pipeline: 16 PUCT simulations, $c_{\text{puct}} = 1.5$, legal-masked priors, root Dirichlet noise $(\alpha, \epsilon) = (0.3, 0.25)$, and temperature 1.0 for 20 plies then 0.1. Each model–seed run has 100 complete mixed-size games and 200 AdamW updates. Complete games contribute root-visit policies and terminal outcomes from the stored player-to-act perspective; truncations never enter replay. Checkpoints at steps 40, 80, 120, 160, and 200 store the optimizer, replay, counters, and RNG states and resume only after hash validation.

5.3 COMPUTE MATCHING AND STATISTICS

The three learned model families match five seeds, training schedule, 16-simulation PUCT budget, optimizer, checkpoints, evaluation schedule, and parameter count (maximum spread 0.6982%). This does not constitute compute matching between a learned model and UCT-MCTS-16. Wall-clock and transitions are recorded but not matched; FLOPs and memory are not yet consolidated. Each final seed-level matchup contains 100 same-seed color-swapped pairs. Pooled Wilson intervals count draws as half a success and are descriptive: the deterministic round robin contains only two action sequences per 200-game task, so no independent-game or multiple-comparison inference is claimed.

6 TOPOLOGY-AWARE REFERENCE MODEL

Each lattice point is a graph node with occupancy, role, coordinate, and initial-node features. Typed edges represent (i) fixed triangular-lattice adjacency, (ii) black logical connections, and (iii) white logical connections. A relation-aware message-passing network produces per-node action logits and a pooled scalar value. A separate global logit represents pass.

All inputs are canonicalized to the player to act. Each node receives 11 features: five occupancy indicators, two normalized coordinates, boundary membership, own and opponent initial-node indicators, and the consecutive-pass count. For relation $r \in \{\text{physical}, \text{own}, \text{opponent}\}$, one message-passing layer for a valid node has the form

$$\begin{aligned} u_i^{(\ell)} &= W_0^{(\ell)} h_i^{(\ell)} + b_0^{(\ell)} + \sum_r W_r^{(\ell)} \frac{1}{\max\{1, |\mathcal{N}_r(i)|\}} \sum_{j \in \mathcal{N}_r(i)} h_j^{(\ell)}, \\ h_i^{(\ell+1)} &= \text{LayerNorm}\Big(h_i^{(\ell)} + \text{ReLU}(u_i^{(\ell)}) \Big). \end{aligned} \tag{3}$$

An empty neighborhood contributes a zero message, and padded nodes are masked after every block and before mean pooling. Grid-GNN exposes only the physical relation; Topology-GNN additionally remaps the two color-specific logical edge sets to own/opponent relations. Padded-CNN maps the same node features to a zero-padded square tensor and does not read any edge relation. All models emit one logit per point, a separate PASS logit, and a `tanh` value; padding and illegal actions are masked.

Table 2: Matched model families from the frozen parameter audit. Counts are within 0.6982%, but architectures, FLOPs, memory, and wall time are not matched.

Model	Width	Parameters	Structural input
Padded-CNN	33	62,868	node features only
Grid-GNN	82	62,733	physical lattice edges
Topology-GNN	64	63,171	physical + typed logical edges

Each network uses three residual convolution/message-passing blocks. Widths were chosen before the formal run to keep the maximum parameter spread below the frozen 1% tolerance. TopologyHistory remains a declared observation variant but was not included in the neural comparison.

7 EXPERIMENTS

7.1 ENVIRONMENT AND DIAGNOSTIC EVIDENCE

The frozen environment/search suite has 62 passing tests; 65 targeted or seeded trajectories match an independent Web implementation over sizes 5–15. Complete side-five enumeration covers 25,096 states and 67,505 transitions. All 1,088 topology-distinct Grid-equivalent pairs have identical legal sets, one-step observations, exact values, shared-action values, and optimal actions. A PASS-aware side-six search finds a natural Superko trigger, while a separate 5,000-pair, 12,656-shared-action bounded continuation search finds no strict paired witness. The latter is a budgeted miss, not an absence proof.

7.2 WITHIN-SIZE SELF-PLAY

The frozen D14–D16 formal ledger contains 15 training and 315 evaluation tasks: 1,500 self-play games, 3,000 updates, and 30,600 task-indexed replay-verified game records, with no failure or truncation. All nine model–size cells beat Random (Table 3). Against UCT-MCTS-16, Grid-GNN at $n = 5$ is inconclusive, Padded-CNN at $n = 5$ is borderline, and the other seven cells have descriptive lower bounds above 0.5.

Table 3: Final-checkpoint score rates on the three training sizes. Each cell pools 1,000 games over five training seeds; brackets are descriptive Wilson 95% intervals recomputed from `final_vs_search.csv`. Score is $(W + 0.5D)/N$, not raw win rate; pooled games are not treated as independent.

Model	n	vs. Random	vs. UCT-MCTS-16
Grid-GNN	5	.851 [.827,.871]	.523 [.492,.553]
	7	.892 [.871,.909]	.646 [.615,.675]
	9	.951 [.936,.963]	.848 [.824,.869]
Padded-CNN	5	.873 [.851,.892]	.532 [.501,.562]
	7	.939 [.922,.952]	.711 [.682,.738]
	9	.978 [.967,.985]	.914 [.895,.929]
Topology-GNN	5	.863 [.840,.883]	.643 [.612,.672]
	7	.910 [.890,.926]	.715 [.686,.742]
	9	.949 [.933,.961]	.876 [.854,.895]

Five checkpoint probes show non-monotone, size-dependent trajectories. Because no area-under-curve or fixed-strength estimator was preregistered, they do not establish relative sample efficiency.

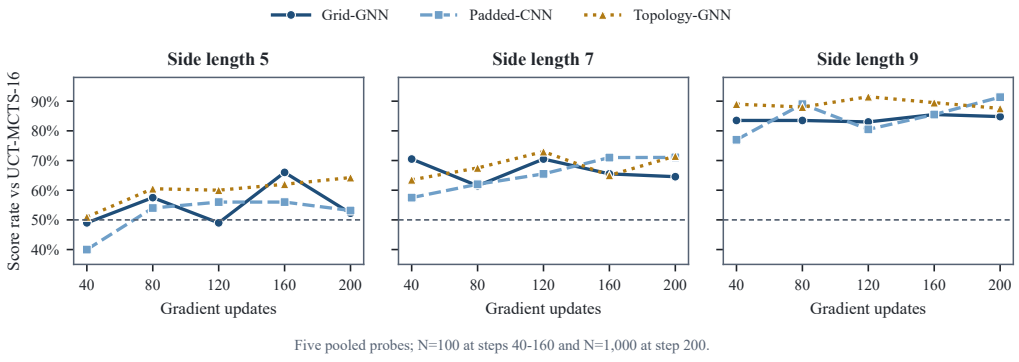


Figure 2: Formal score-rate probes against UCT-MCTS-16 from `arena_learning_curve.csv`. The vertical scale is focused on 0.35–0.98. Steps 40–160 pool 100 games per point, whereas step 200 pools 1,000; the five discrete probes support neither a continuous-convergence nor a relative sample-efficiency claim.

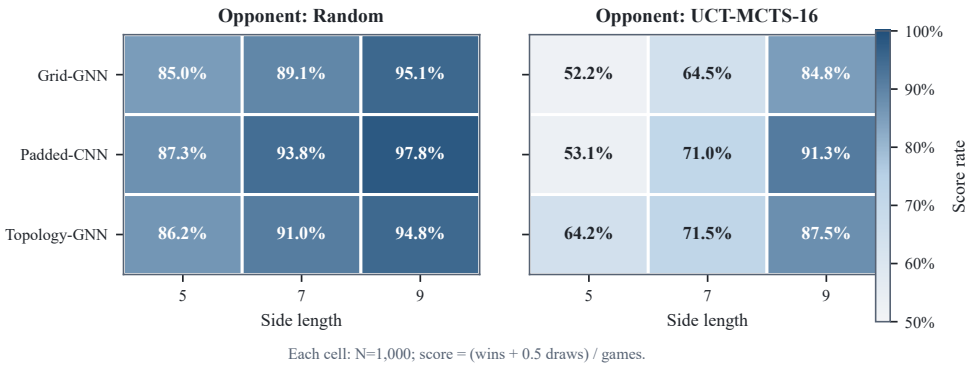


Figure 3: Final model–size–opponent score rates from `final_vs_search.csv`; each cell pools 1,000 games from five training-seed receipts. Score is $(W + 0.5D)/N$, not raw win rate. Colors do not encode uncertainty, multiplicity-adjusted significance, or unseen-size generalization.

7.3 Representation round robin

Direct play exhibits a size–representation interaction (Table 4). Topology-GNN outscores Grid-GNN at all three sizes, while Padded-CNN reverses direction against both graph models. No global ordering is supported.

Table 4: Score from A's perspective in `representation_round_robin.csv`, pooled over five matched training seeds. Intervals are omitted because each 200-game task repeats two deterministic color-conditioned trajectories; no global ranking is inferred.

A vs. B	$n = 5$	$n = 7$	$n = 9$
Topology-GNN vs. Grid-GNN	.65	.80	.60
Padded-CNN vs. Grid-GNN	.50	.55	.30
Padded-CNN vs. Topology-GNN	.60	.30	.50

Across all 9,000 round-robin games, each 200-game task contains exactly two unique color-conditioned trajectories. Each aggregate therefore represents five training seeds and ten unique trajectories, not 1,000 independent trials.

7.4 Ablations

`enforce` rejects repetitions, whereas `observe` admits them while retaining history. In a 2,000-pair fixed-policy pilot at each of sizes 6 and 7, only two size-7 games triggered Superko and one trajectory diverged; all winners agreed, no game truncated, and mean length changed by -0.01017%.

All internal fixed gates for score (± 5 percentage points), relative length ($\pm 5\%$), and truncation passed for this policy, size set, and horizon. This supports "empirically similar under the sampled policy," not rule equivalence: only one handcrafted policy was evaluated.

A preregistered five-seed extension covered six policies, two sizes, 120 matched pairs, and 240 replay-verified games. It observed no trigger or outcome effect, but every practical-similarity gate failed: each cell had only ten `observe` games, so $0/10$ has a 27.75% Wilson upper bound, above the 1% threshold. CycleSeeking's null result is a bounded search miss, and its neural row is a one-game, one-update smoke checkpoint.

7.5 Compute and reproducibility

Runs used Python 3.10.20, PyTorch 2.7.1+cu118, deterministic CUDA 11.8, and an RTX 4060 Laptop GPU. Mean per-seed training time was 489.8 s for Padded-CNN, 399.2 s for Grid-GNN, and 440.0 s for Topology-GNN (1.85 aggregate training GPU-hours); evaluation compute is not

yet consolidated. One canonical manifest digest (`1bbf2c9a9ecc...`) binds all 330 receipts. Independent aggregation rechecked task identity, budgets, hashes, summaries, and all 30,600 task-indexed replay-verified game records; for this completed D14–D16 bundle, `formal_ready=true` denotes artifact completeness, not superiority.

8 Limitations and Broader Impact

- Results cover jointly trained sizes 5, 7, and 9, not zero-shot sizes 10 or 12; history, logical-edge, cascade, and search-budget ablations remain.
- Grid-GNN and Topology-GNN differ in typed relations and hidden width, so their comparison mixes representation and architecture; the CNN adds a model family confound. Parameters and data are matched, but FLOPs, memory, transitions, and wall time are not.
- Each D16 200-game task contains only two unique color-conditioned trajectories. These repeats invalidate independent-game uncertainty; a later preregistered diversified extension executed all planned games but failed its frozen trajectory-diversity gate and is therefore excluded here. A prospectively redesigned evaluation and independent verification remain required.
- Lifeline-RL is one hand-designed game, so external validity to other games is untested; its Superko matrix is finite-policy evidence. The smoke neural condition and incomplete evaluation-compute accounting preclude broader efficiency claims.

9 Conclusion

Lifeline-RL isolates dynamic topology and history-dependent legality in a replayable benchmark. Side-five enumeration yields a decision-relevant negative result, while a side-six trace proves Superko can bind. Under one frozen five-seed protocol, all three learned representations beat Random on seen sizes; Topology-GNN outscores Grid-GNN directly, but size-dependent reversals and duplicated trajectories rule out a global ranking. Next steps are a prospectively frozen paired-opening evaluation, independent zero-shot verification, and only then gated rule/feature ablations.

Reproducibility statement

Sections 5–7 and Appendix A specify the frozen protocol, model budgets, statistics, and result boundaries. The accompanying claim ledger maps each paper claim to compact aggregate CSV/JSON, append-only receipt-ledger JSONL, and independent verifier outputs. Checkpoints, raw per-game logs, and local task directories are not claimed as submission artifacts.

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A Hyperparameters and Training Details

Table 5: Frozen formal settings from `D14_D16_EXPERIMENT_PROTOCOL.md`, shared by all three learned representations. They define the completed seen-size run, not an unseen-size or search-budget ablation.

Setting	Value
Training sizes / seeds	$\{5, 7, 9\}$ / 20260825–20260829
PUCT	16 simulations, $c_{\text{puct}} = 1.5$
Root noise	$\alpha = 0.3$, $\epsilon = 0.25$
Temperature	1.0 for 20 plies, then 0.1
Self-play / updates	100 complete games / 200 gradient steps
Replay / batch	10,000 / 16
Optimizer	AdamW, lr 10^{-3}, weight decay 10^{-4}
Gradient clipping	global norm 5.0
Checkpoints	steps 40, 80, 120, 160, 200
Terminal rule	256 plies; enforced Superko; no shaped reward
Final evaluation	200 games per seed and matchup
Color control	100 same-seed swapped-color pairs

Targets are the masked root-visit distribution and the terminal result from the stored player-to-act perspective. Absolute player payoffs are backed up through the search tree because automatic skips can leave the same player to act after a transition. Training loss is masked policy cross-entropy plus mean-squared terminal value error.

B Additional Results and Reproducibility

Canonical aggregates retain checkpoint, W/D/L, seed, loss, transition, and timing fields. In the completed D14–D16 bundle, no formal task failed or truncated; smoke runs and duplicate views of the step-200 UCT matchup are excluded from the 30,600 task-indexed replay-verified game records. The paper's evidence mapping names frozen protocol/config hashes, compact aggregate CSV/JSON summaries, an append-only JSONL receipt ledger, and independent verifier outputs. It does not claim that checkpoints, raw per-game logs, or task directories are included in the submission artifacts. Aggregation rejects missing or duplicate tasks, identity or budget drift, truncation, color imbalance, and incomplete replay. Two interrupted runners lack independent canonical files for their first two attempts, so no per-attempt lineage is claimed.